

# Solving Logarithmic Equations

Algebra Worksheet · Grade 10–12

Name: \_\_\_\_\_

Date: \_\_\_\_\_

## Learning Objectives

- Apply logarithm properties (power, product, quotient) to simplify logarithmic expressions
- Solve logarithmic equations by canceling logarithms when both sides share the same base
- Convert between logarithmic and exponential form to solve equations leading to linear or quadratic solutions

## Problems

1. Solve the logarithmic equation by canceling the logarithms on both sides.

$$\log_b (x + 3) = \log_b (7)$$

2. Solve the logarithmic equation by canceling the logarithms on both sides.

$$\log_5 (2x - 1) = \log_5 (9)$$

3. Use the power property of logarithms to rewrite the left side, then solve.

$$3\log_b (x) = \log_b (27)$$

4. Use the product property of logarithms to combine the left side, then convert to exponential form and solve.

$$\log_2 (x) + \log_2 (4) = 3$$

5. Solve the equation. Use the cancellation property after matching both sides.

$$\log_b (3x + 4) = \log_b (5x - 2)$$

6. Use the product property to combine the left side, convert to exponential form, and solve the resulting equation.



$$\log_3(x) + \log_3(x - 2) = 1$$

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7. Use the power property on the left side and the product property on the right side, then solve.

$$2\log_b(x) = \log_b(3) + \log_b(x + 6)$$

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8. Combine using the quotient property on the left side, convert to exponential form, and solve the quadratic.

$$\log_2(x + 5) - \log_2(x - 1) = 2$$

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9. Use logarithm properties to simplify both sides, then solve the resulting quadratic equation.

$$2\log_b(x) = \log_b(4) + \log_b(x - 1)$$

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10. Simplify using logarithm properties on both sides, convert to exponential form if needed, and solve the resulting quadratic equation.

$$\log_2(x) + \log_2(x - 2) = \log_2(x + 4)$$

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# Solving Logarithmic Equations — Answer Key

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## Answer Key

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### 1. Answer: $x = 4$

- Since both sides have the same base  $b$ , cancel the logarithms:  $x + 3 = 7$
- Subtract 3 from both sides:  $x = 4$

### 2. Answer: $x = 5$

- Cancel the logarithms (same base 5):  $2x - 1 = 9$
- Add 1 to both sides:  $2x = 10$ , then divide by 2:  $x = 5$

### 3. Answer: $x = 3$

- Apply power property:  $3 \log_b(x) = \log_b(x^3)$
- Cancel logarithms:  $x^3 = 27$
- Take the cube root:  $x = 3$

### 4. Answer: $x = 2$

- Apply product property:  $\log_2(4x) = 3$
- Convert to exponential form:  $4x = 2^3 = 8$
- Divide both sides by 4:  $x = 2$

### 5. Answer: $x = 3$

- Cancel the logarithms (same base):  $3x + 4 = 5x - 2$
- Subtract  $3x$  from both sides:  $4 = 2x - 2$
- Add 2:  $6 = 2x$ , so  $x = 3$

### 6. Answer: $x = 3$

- Apply product property:  $\log_3(x(x-2)) = 1$
- Convert to exponential:  $x(x-2) = 3^1 = 3$
- Expand:  $x^2 - 2x - 3 = 0$
- Factor:  $(x-3)(x+1) = 0$ , so  $x = 3$  or  $x = -1$
- Reject  $x = -1$  (negative argument), so  $x = 3$

### 7. Answer: $x = 6$

- Power property:  $\log_b(x^2) = \log_b(3(x+6))$
- Cancel logarithms:  $x^2 = 3x + 18$
- Rearrange:  $x^2 - 3x - 18 = 0$
- Factor:  $(x-6)(x+3) = 0$ , so  $x = 6$  or  $x = -3$
- Reject  $x = -3$  (negative argument), so  $x = 6$

### 8. Answer: $x = 3$

- Apply quotient property:  $\log_2((x+5)/(x-1)) = 2$





- Convert to exponential:  $(x+5)/(x-1) = 4$
  - Multiply both sides by  $(x-1)$ :  $x + 5 = 4x - 4$
  - Solve:  $9 = 3x$ , so  $x = 3$
  - Check:  $x = 3$  gives valid (positive) arguments
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**9. Answer:  $x = 2$** 

- Power property left side:  $\log_b(x^2)$ ; product property right side:  $\log_b(4(x-1))$
  - Cancel logarithms:  $x^2 = 4x - 4$
  - Rearrange:  $x^2 - 4x + 4 = 0$
  - Factor:  $(x-2)^2 = 0$ , so  $x = 2$
  - Check:  $x = 2$  gives valid arguments in original equation
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**10. Answer:  $x = 4$** 

- Product property on left:  $\log_2(x(x-2)) = \log_2(x+4)$
  - Cancel logarithms:  $x(x-2) = x + 4$
  - Expand:  $x^2 - 2x = x + 4$
  - Rearrange:  $x^2 - 3x - 4 = 0$
  - Factor:  $(x-4)(x+1) = 0$ , so  $x = 4$  or  $x = -1$
  - Reject  $x = -1$  (negative argument), so  $x = 4$
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